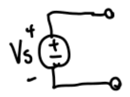
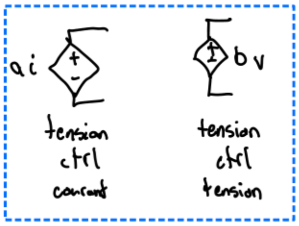
Rappel

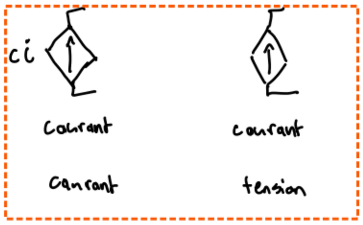
$q = 1{,}6022 \times 10^{-19}$ C

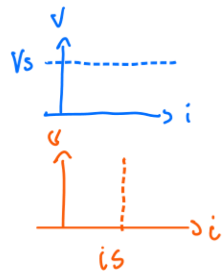
$v = \frac{dw}{dq}$ $\quad$ $i = \frac{dq}{dt}$

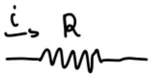
$p = +iv$ ou $-iv$



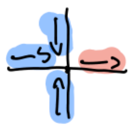




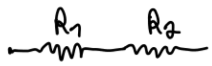




$V = Ri$ $\quad$ $p = Ri^2 = \frac{v^2}{R}$ $\quad$ $G = 1/R$

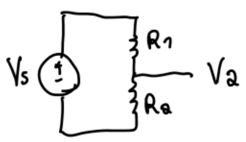


$i_{in} = i_{out}$

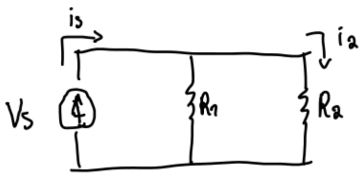


$R = R_1 + R_2$

$R = (1/R_1 + 1/R_2)^{-1}$

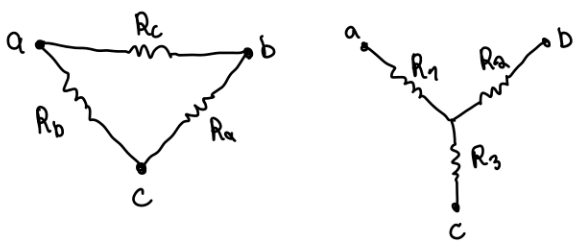


$V_2 = V_s \cdot \frac{R_2}{R_1 + R_2}$



$i_2 = i_s \cdot \frac{R_1}{R_1 + R_2}$

Trans phorm Δ Y



$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{ac} = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} = R_1 + R_3$$

$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$\Delta \rightarrow Y$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$Y \rightarrow \Delta$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$